

BreakoutTrendAI

Self-Learning Neural Network EA for MetaTrader 5

VERSION
v3.0

DATE
June 2026

TIMEFRAME
M15

PAIRS
6 Pairs

PROFILE
Prop Firm (FTMO / GoatFunded / MFF)

1 Overview

BreakoutTrendAI is a fully autonomous, self-learning Expert Advisor for MetaTrader 5. It combines a classical 20-bar **range breakout strategy** with an **online Multi-Layer Perceptron (MLP)** that improves after every closed trade. Designed specifically for prop firm challenges — it has built-in daily/weekly loss limits, session filters, and a SafeGuard circuit breaker.

PROPERTY	DETAIL
Strategy type	Range breakout + trend filter (H1 EMA50) + AI score filter
AI model	MLP 32 → 24 → 12 → 1 with ReLU activations, SGD + Momentum backprop
Learning trigger	After every closed trade (profit/loss → label → backprop)
Timeframe	M15 (signals on M15 bars, trend filter on H1)
Trading pairs	EURUSD · GBPUSD · XAUUSD · USDJPY · GBPJPY · AUDUSD
Risk model	ATR-based position sizing, fixed % risk per trade
Prop firm safe	Daily loss limit, weekly loss limit, max trades/day, SafeGuard EA link
File	MQL5\Experts\BreakoutTrendAI_Single.mq5

2 How It Works — Trade Logic Flow



Key design principle: Every filter must pass before a trade opens. If any step fails, the dashboard shows exactly why (OUT OF SESSION / NO TREND / NO BREAKOUT / SCORE LOW / etc.)

Trend Filter (Step 4)

DIRECTION	CONDITION
BUY signal	M15 close > H1 EMA50 AND RSI(14) > 45
SELL signal	M15 close < H1 EMA50 AND RSI(14) < 55

Range Breakout (Step 5)

Range = highest high / lowest low of bars **2 to 21** (bar 1 excluded so bar 1's close can actually break out). Range must be $0.5 \times \text{ATR}$ – $5.0 \times \text{ATR}$ wide (ATR-based filter replaces fixed pip filter).

Position Management

FEATURE	BEHAVIOUR
Stop Loss	$1.5\text{--}2.0 \times \text{ATR}(14)$ from entry (pair-dependent)
Take Profit 1	$1.0 \times \text{SL distance}$ (1:1 RR) — partial target
Take Profit 2	$2.5 \times \text{SL distance}$ (2.5:1 RR) — full target
Break-even	SL moved to entry when TP1 is hit
Max 1 trade per pair	EA skips if position already open on same symbol + magic

3 AI Self-Learning Engine

Neural Network Architecture

LAYER	NEURONS	ACTIVATION
Input	32 features	—
Hidden 1	24	ReLU
Hidden 2	12	ReLU
Output	1 (score 0–1)	Sigmoid

32 Input Features

#	FEATURE	#	FEATURE
0	ATR / price (volatility)	16	Body size bar 7
1	Spread / ATR	17–20	High-low range bars 1–4 / ATR
2	Range width / ATR	21	Upper wick bar 1 / ATR
3	Price vs EMA50 / ATR	22	Lower wick bar 1 / ATR
4	Price vs EMA200 / ATR	23	Entry distance from range high / ATR
5	EMA50 vs EMA200 / ATR	24	Entry distance from range low / ATR
6	RSI(14) / 100	25	10-bar momentum / ATR
7	Signal direction (+1/–1)	26	Distance to swing high / ATR
8	Hour of day / 23	27	Distance to swing low / ATR
9	Day of week / 6	28	Bollinger Band width / ATR (squeeze)
10–16	Body size bars 1–7 / ATR	29	H1 EMA50 direction (+1/–1)
—	—	30	Distance to prior day high / ATR
—	—	31	Distance to prior day low / ATR

Learning Cycle

When a trade closes:

1. Profit (incl. swap + commission) is converted to a label using: `label = sigmoid(profit% × 20)`
2. Backpropagation updates weights using SGD + Momentum (lr=0.001, momentum=0.90)
3. Model saved to `btai_<symbol>.dat` every 5 trades

Bootstrap Mode

First 10 trades per pair: threshold lowered to **0.40** (from 0.54–0.60) so the AI collects real trade outcomes to learn from. After 10 trades, the full threshold applies.

Model Files (per pair, isolated)

PAIR	MODEL FILE	LOG FILE
EURUSD	btai_eurUSD.dat	btai_eurUSD_log.csv
GBPUSD	btai_gbpUSD.dat	btai_gbpUSD_log.csv
XAUUSD	btai_xauUSD.dat	btai_xauUSD_log.csv
USDJPY	btai_usdJPY.dat	btai_usdJPY_log.csv
GBPJPY	btai_gbpJPY.dat	btai_gbpJPY_log.csv
AUDUSD	btai_audUSD.dat	btai_audUSD_log.csv

Location: C:\Users\<you>\AppData\Roaming\MetaQuotes\Terminal\<id>\MQL5\Files\
Access via MT5: Tools → Data Folder → MQL5 → Files

4 Trading Sessions (UTC)



● London 07:00–13:00 UTC ● London/NY Overlap 13:00–17:00 UTC (highest volume)
● Late New York 17:00–21:00 UTC ● Market Closed (no trades)

SESSION	HOURS (UTC)	TOGGLE INPUT	DEFAULT	BEST FOR
London	07:00 – 13:00	InpTradeLondon	ON	EUR, GBP, Gold
Overlap	13:00 – 17:00	InpTradeOverlap	ON	All pairs — highest volume
Late NY	17:00 – 21:00	InpTradeLateNY	ON	USD pairs
Asia	00:00 – 09:00	InpTradeAsia	OFF	JPY, AUD (optional)

5 Pair Settings & Live Performance

Live account data (GoatFunded Phase 1 — 414572704) used to tune thresholds. Profit Factors shown are from real trades on the account.

AUDUSD BEST

Live Profit Factor: **14.78**

Magic Number	787876
Risk / Trade	0.50%
AI Threshold	0.54
SL Multiplier	1.5× ATR
Max Trades/Day	4
Model File	btai_audusd.dat

EURUSD FILTERED

Live Profit Factor: **0.30** → threshold raised

Magic Number	787871
Risk / Trade	0.50%
AI Threshold	0.55 (was 0.58)
SL Multiplier	1.5× ATR
Max Trades/Day	4
Model File	btai_eurusd.dat

GBPUSD

Live Profit Factor: **1.24**

Magic Number	787872
Risk / Trade	0.40%
AI Threshold	0.57
SL Multiplier	1.8× ATR
Max Trades/Day	4
Model File	btai_gbpusd.dat

XAUUSD HIGH ATR

Live Profit Factor: **0.60** → threshold raised

Magic Number	787873
Risk / Trade	0.35%
AI Threshold	0.60
SL Multiplier	2.0× ATR
Max Trades/Day	3
Model File	btai_xauusd.dat

USDJPY

Tight spreads, clean trends

Magic Number	787874
Risk / Trade	0.45%
AI Threshold	0.55
SL Multiplier	1.5× ATR
Max Trades/Day	4
Model File	btai_usdjpy.dat

GBPJPY VOLATILE

Large ATR spikes — tight risk

Magic Number	787875
Risk / Trade	0.35%
AI Threshold	0.58
SL Multiplier	2.0× ATR
Max Trades/Day	3
Model File	btai_gbpjpy.dat

6 Risk Management

RULE

DEFAULT

BEHAVIOUR

Daily loss limit	3.50%	Stops all new trades for the day
Weekly loss limit	6.00%	Stops all new trades for the week
Max trades / day	3–4 (per pair)	Counter resets at 00:00 UTC
Daily reset	00:00 UTC	Trade counter + daily equity baseline reset automatically
Weekly reset	Monday 00:00 UTC	Weekly equity baseline reset
Position sizing	ATR-based	$\text{Lot} = (\text{Equity} \times \text{Risk}\%) \div (\text{SL pips} \times \text{pip value})$
Min stop distance	Auto	Checked against SYMBOL_TRADE_STOPS_LEVEL before each order
Fill mode	Auto-detect	FOK → IOC → RETURN (broker-compatible, no silent rejections)

GoatFunded / FTMO limits: Daily 5% · Total 10%. BreakoutTrendAI uses 3.5% / 6% as internal limits — this gives a 1.5% safety buffer so the prop firm limit is never accidentally hit by a single bad candle.

7 SafeGuard EA — Account Circuit Breaker

A separate EA ([MQL5\Experts\SafeGuard_EA.mq5](#)) that runs on **one chart only** and monitors the entire account. When a trigger fires, it closes ALL open positions and locks BreakoutTrendAI via a GlobalVariable for a set period.

TRIGGER	DEFAULT	ACTION
Dollar loss limit	\$500 (range: \$100–\$1,200)	Close all + lock for Pause Hours
Drawdown %	3.0% of account	Close all + lock for Pause Hours
Consecutive losses	3 in a row	Close all + lock for Pause Hours
Pause duration	4 hours	Auto-unlocks after timer expires
Daily reset	00:00 UTC	Loss counters reset, lock cleared

Communication method: SafeGuard writes a GlobalVariable

SAFEGUARD_<AccountLogin> = expiry timestamp. BreakoutTrendAI reads this on every bar
— if current time < expiry, all entries are blocked.

8 Installation & Setup

Step 1 — Copy Files to MT5

FILE	DESTINATION (MT5 DATA FOLDER)
BreakoutTrendAI_Single.mq5	MQL5\Experts\
SafeGuard_EA.mq5	MQL5\Experts\
BTAI_EURUSD_M15.set	MQL5\Presets\
BTAI_GBPUSD_M15.set	MQL5\Presets\
BTAI_XAUUSD_M15.set	MQL5\Presets\
BTAI_USDJPY_M15.set	MQL5\Presets\
BTAI_GBPJPY_M15.set	MQL5\Presets\
BTAI_AUDUSD_M15.set	MQL5\Presets\

Step 2 — Compile in MetaEditor

1. Open MetaEditor (press F4 in MT5)
2. Open BreakoutTrendAI_Single.mq5
3. Press F7 to compile → must show 0 errors
4. Repeat for SafeGuard_EA.mq5

Step 3 — Attach to Charts

Open 6 × M15 charts: EURUSD · GBPUSD · XAUUSD · USDJPY · GBPJPY · AUDUSD
For each chart:

- Drag BreakoutTrendAI_Single from Navigator → Expert Advisors
- Properties → Load (bottom left) → select the matching .set file
- Click OK

Open 1 extra chart (any symbol):

- Attach SafeGuard_EA (no preset needed)
- Set Inp_MaxDollarLoss to your risk tolerance (\$100–\$1,200)

Step 4 — Verify

Dashboard should show: ● **RUNNING** · Session name · Train Steps: 0 (BOOTSTRAP)
After the first closed trade on each pair, check MT5 Files folder — you should see **6 separate .dat files** appearing (one per pair).

9 Dashboard — Reading the Panel

FIELD	MEANING	GOOD VALUE
STATUS	EA running state	● RUNNING
SESSION	Current active session	● LONDON / ● OVERLAP / ● LATE NY
Train Steps	Number of completed learning cycles	Growing over time. 0–9 = BOOTSTRAP
Last Score	AI output for most recent signal (0–1)	≥ threshold = trade opens
Threshold	Minimum score to open a trade	Pair-dependent (0.54–0.60)
Last Signal	BUY ▲ or SELL ▼	Matches trend direction
Reason	Why last bar did or did not trade	See table below
Daily Loss	Current day loss vs limit	Green = safe, Red = limit close
Trades Today	Count vs max allowed	e.g. 2 / 4

Reason Codes

REASON DISPLAYED	MEANING
OUT OF SESSION	Current UTC hour is outside enabled sessions
LIMIT HIT	Daily loss / weekly loss / max trades reached
SAFEGUARD LOCKED	SafeGuard EA triggered — wait for timer to expire
TRADE OPEN	Position already open on this pair (max 1)
NO TREND (H1 EMA50)	Price not clearly above/below H1 EMA50
NO BREAKOUT	Last bar close did not break the 20-bar range
BOOTSTRAP x.xxx	AI <10 trades, score shown, bootstrap threshold used
SCORE LOW x.xxx	AI score below threshold — signal filtered
STOPS TOO CLOSE	SL/TP within broker's minimum stop level
OPENED #xxxxxx	Trade opened successfully — ticket number shown
ERR 10030: ...	Broker rejected order — error code + reason shown

10 Version History

VERSION	DATE	KEY CHANGES
v1.0–2.1	May 2026	Initial build. Score always 0 (range filter in wrong units, EMA50/200 cross never triggered)

v2.2	Jun 2026	Fixed trend filter: M15 EMA50/200 cross → H1 EMA50 comparison
v2.3	Jun 2026	Fixed range breakout: iHighest start=1 → start=2 (breakout was mathematically impossible)
v2.4	Jun 2026	Added bootstrap mode (threshold 0.40 for first 10 trades)
v2.5	Jun 2026	Added Asian session toggle
v2.6	Jun 2026	Added SafeGuard link via GlobalVariable
v2.7	Jun 2026	Auto-detect broker fill mode (FOK → IOC → RETURN). Min stop level check. Error in dashboard.
v2.8	Jun 2026	Sessions split: London / Overlap / Late NY — each independently toggleable
v2.9	Jun 2026	Fixed daily reset — tradesToday never reset at midnight (critical bug causing LIMIT HIT all day)
v3.0	Jun 2026	Auto-name model file per symbol — all 6 pairs no longer share btai_model.dat